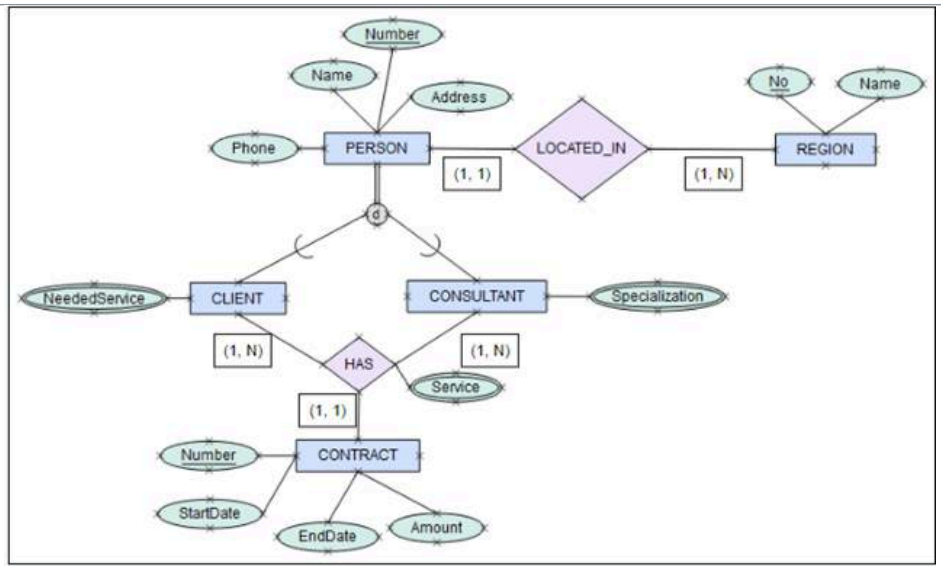
Question 1 [5 marks] Place the appropriate structural constraint onto each blank space of the EER diagram below, according to the specification given in the panel above.



Question 2 There are multiple solutions possible

PERSON (Number, Name, Address, Phone, Region, Client, Consultant)

Region (No, Name)

ClientServices (Client, Service)

ConsultantSpecialization (Consultant, Specialization)

Contract (Number, StartDate, EndDate, Amount)

ContractBetween (Contract, Client, Consultant)

ContractedServices (Contract, Client, Consultant, Service)

Question 3

Using a file instead of a database has many consequences. There is no automatic security checking, back up and access control. There is no automatic checking of input data, as is the case in a database system. If multiple users need to modify the data, there might be access problems, and problems maintaining data consistency.

The Excel file combines attributes of various entity types: Client, Region, Consultant, Contract. Viewed from the point of view of the relational data model, such table is ill-designed, and will suffer from insertion, deletion and modification anomalies. Furthermore, there are multiple values for some of the cells in the sheet – e.g. specialization for consultants. The file might be very large, which will make it difficult to search for needed information and understand the data. There might be a lot of redundant data.

Insertion anomaly: in order to add a new contract for an existing client, it is necessary to add the client info again – potentially leading to mistakes.

Example for a deletion anomaly: If the last contract for a particular client is deleted, there is no information about the client anymore.

Modification anomaly: if we want to add specialization for a consultant, multiple rows might need to be modified, potentially leading to mistakes.

Question 4

?

EMPLOYEE: Primary key is Emp_No. No foreign keys

PILOT: Primary key is Emp_No. Foreign keys - EmpNo

FLIGHT: Primary key is FlightNo. Foreign keys: Pilot, Copilot

Question 5

Select FlightNo

From flight F1

Where (Pilot = 105 or copilot = 105) and

not exists (select * from flight F2

where F2.FlightNo = F1.FlightNO and

(Pilot = 109 or copilot = 109));

The alternative is to use two queries with set difference.

$\pi_{FlightNo}\ (\sigma_{Pilot = 105\ OR\ CoPilot = 105}\ (Flight)) - \pi_{FlightNo}\ (\sigma_{Pilot = 109\ OR\ CoPilot = 109}\ (Flight))$

Question 6

a. $\pi_{Pilot}\ (Flight)$

Show the employee number for each pilot in the Flight table.

b. $\pi_{FName,LName,Hire_Date,Medical_Certificate,Licence_Date}\ ((\sigma_{Flight=1003}\ (Flight)) \bowtie_{Copilot = EmpNo}\ Pilot) \bowtie_{Pilot.EmpNo = Employee.EmpNo}\ (EMPLOYEE)$

Show names, hire dates, and the dates of medical certificates for the copilot of flight 1003.

c. $_{Pilot}\mathcal{F}_{COUNT\ FlightNo}\ (Flight)$

For all pilots in the Flight table, show how many flights they have.

Question 7

a. Key and superkey

Both keys and superkeys are sets of attributes which have unique values within the table. The difference is that the key is minimal: no attributes can be removed from the key and still have unique values. Every key is at the same time a superkey.

b. Disjoint and overlapping specialization

An entity belonging to the superclass can belong to only one of the subclasses in the case of a disjoint specialization. For overlapping specialization, an entity from the superclass may belong to more than one subclass.

c. Total and partial category

The total category means that each entity belonging to the superclasses must belong to the category. That is not the case for the partial categories, where there could be some entities from the superclasses that do not appear in the category.

d. Conceptual and logical database design

Conceptual database design is done in a high-level model such as EER, and represents the content of the database in terms of entities, attributes and relationships. Logical database design is performed using an implementational data model, such as the relational data model, and such a schema can be implemented in a DBMS.

e. Views and base tables

> Views are defined by specifying a Select statement, which is then stored in the data catalog (together with the name of the view, and the attributes names). When the view is used in a query, the defining Select statement is executed first.
>
> Base tables are actually stored on the physical level (in files).

Question 8

> The integrity component consists of four constraints which ensure that the database is always in a consistent state. The constraints are:
>
> - Domain constraint: this constraint check that when a new value is added (or existing values are updated), the value is from the domain (correct type and satisfies any conditions specified for the attribute).
> - Key constraint: every table has a unique identifiers. This includes the primary key and any candidate keys (Primary Key, Unique)
> - Entity-integrity constraint: no component of the primary key can be null.
> - Referential integrity constraint: the value of the foreign key must match the value of corresponding attribute(s in the referenced table, or it can be NUL.

Question 9 [10 marks] Relation R(A, B, C, D, E, G, H) is given, as well as a set of functional dependencies:

F = {AB → C, DEG → H, A → B, DG → H}

Find the minimal cover for F.

> The minimal cover is { A → B, A → C, DG → H}

Question 10 [2 marks]

> Only one clustered index can be created on a table.

Question 11 [2 marks]

> $\log_2(250{,}000)$

Question 12 [4 marks]

> 1. Index with one entry for each table row: Dense index
> 2. Index with one entry for each table block: Sparse index
> 3. An index on the table sort column: Primary index
> 4. An index that is not on the table sort column: Secondary index

Question 13 [5 marks]

> A, B, D, F, K

Question 14 [3 marks]

> Dirty Read

Question 15 [3 marks]

> Nonrepeatable Read

Question 16 [3 marks]

> Phantom Read

Question 17 [2 marks]

Indexing

Sorting

Hashing

Question 18 [1 mark]

False

Question 19 [5 marks]

1. B
2. D
3. No

◄ Exam 2023

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